

Numerical integration

$$\int_a^b f(x) dx$$

$$\int_{\Omega} f(x) d\mu, \quad \Omega \subseteq \mathbb{R}^d$$

- ▶ Analytical integration (using indefinite integral)
- ▶ Numerical integration (using optimal numerical rules)
- ▶ Monte Carlo Markov chains (MCMC) (using statistical methods)
Focus of these part of the course!

Analytical integration

- ▶ If you know the indefinite integral, then it is **exact** and thus **best** of all three methods.
- ▶ Beyond cases of elementary functions, indefinite integral might be **difficult to derive** or the explicit formula does **not even exist**:

$$\underbrace{\int e^{-x^2} dx}_{\text{in normal distribution}}, \quad \underbrace{\int \sqrt{1 - \frac{1}{2} \sin^2 x} dx}_{\text{in arc length of an ellipse}}$$

Numerical integration

- ▶ In **low dimensions** and with **nice Ω** much better than MCMC in terms of **computational burden for the same error estimates**.
- ▶ Their forms:

$$\int_{\Omega} f d\mu \approx \rho_1 f(x_1) + \rho_2 f(x_2) + \dots + \rho_m f(x_m) \quad (1)$$

$\rho_i \in [0, \infty)$... weights or densities

$x_i \in \Omega$... nodes or atoms

- ▶ Procedure:
Find **lowest possible m** such that the formula (1) is **exact (without error)** for **polynomials up to certain degree**.

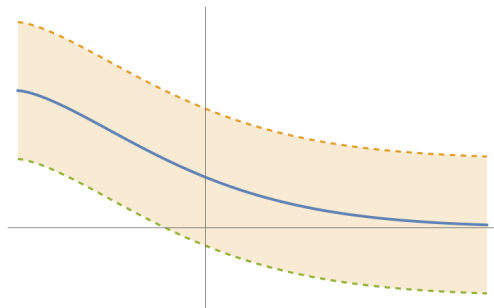
Determine ρ_i and x_i .

Why exactness for polynomials up to certain degree?

Theorem (Stone-Weierstrass theorem, 1885)

Let $X \subseteq \mathbb{R}^d$ be a compact set and $C(X, \mathbb{R})$ the set of continuous functions on X . For every $f \in C(X, \mathbb{R})$ and every $\epsilon > 0$ there exists a polynomial $p \in \mathbb{R}[x]$ such that

$$\|f - p\|_{\infty} = \sup_{x \in X} |f(x) - p(x)| < \epsilon.$$



Numerical integration

One-dimensional case

Input: A degree n sequence $\beta_0, \dots, \beta_n \in \mathbb{R}$.

Output: Atoms $x_1, \dots, x_m$ and densities $\rho_1, \dots, \rho_m$ such that

$$\beta_i = \rho_1 x_1^i + \rho_2 x_2^i + \dots + \rho_m x_m^i.$$

Example - integration w.r.t. the Lebesgue measure on $[a, b]$:

$$\begin{aligned}\beta_0 &= \int_a^b 1 dx = b - a = \rho_1 + \rho_2 + \dots + \rho_m, \\ \beta_1 &= \int_a^b x dx = \frac{b^2 - a^2}{2} = \rho_1 x_1 + \rho_2 x_2 + \dots + \rho_m x_m, \\ &\vdots \\ \beta_n &= \int_a^b x^n dx = \frac{b^{n+1} - a^{n+1}}{n+1} = \rho_1 x_1^n + \rho_2 x_2^n + \dots + \rho_m x_m^n.\end{aligned}\tag{2}$$

The **system (2)** is nonlinear (in ρ_i, x_i) and thus difficult to solve.

Numerical integration

Question 1: Is the system from the previous slide always solvable? ... If $m \geq n + 1$, yes:

- ▶ Let $x_1, \dots, x_{n+1}$ form the division of the interval $[a, b]$ into n intervals of equal length, i.e., $\frac{b-a}{n}$.



- ▶ Form a Vandermondo matrix

$$V = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_{n+1} \\ \vdots & & & \vdots \\ x_1^n & x_2^n & \cdots & x_{n+1}^n \end{pmatrix}, \quad \det V = \prod_{i \neq j} (x_i - x_j).$$

- ▶ Solve the sistem $V\rho = \beta$, where

$$\rho = (\rho_1 \quad \rho_2 \quad \cdots \quad \rho_{n+1})^T, \quad \beta = (\beta_0 \quad \beta_2 \quad \cdots \quad \beta_n)^T,$$

Numerical integration

Question 2: What is the error of the formula above?

- Compute the next element in the sequence,

$$\beta_{n+1} = \int_a^b x^{n+1} d\mu$$

and determine the constant C in the error expression:

$$\beta_{n+1} - (\rho_1 x_1^{n+1} + \dots + \rho_{n+1} x_{n+1}^{n+1}) = \frac{C}{(n+1)!} \left(\frac{b-a}{n} \right)^{n+2}$$

- Why error is of this form?

This follows by integrating the error in each point when approximating a function with a polynomial and requires some work to derive.

- What about the error for an arbitrary f ?

In this case C is equal to $Df^{(n+1)}(\xi)$, where D is a constant and $\xi \in [a, b]$.

Numerical integration

Question 3: Is choosing equidistant points $x_1, \dots, x_{n+1}$ on $[a, b]$ optimal? ... No:

- ▶ In this case we lose $n + 1$ degrees of freedom. Only $n + 1$ densities ρ_i can be optimized.
- ▶ Allowing also the atoms $x_1, \dots, x_{n+1}$ to be free, we may also pose further $n + 1$ conditions

$$\beta_i = \rho_1 x_1^i + \rho_2 x_2^i + \dots + \rho_m x_m^i, \quad i = n + 1, \dots, 2n + 1.$$

- ▶ The error term

$$\beta_{2n+2} - (\rho_1 x_1^{2n+2} + \dots + \rho_{n+1} x_{n+1}^{2n+2}) = \frac{C}{(2n+2)!} \left(\frac{b-a}{n} \right)^{2n+3}$$

will be smaller.

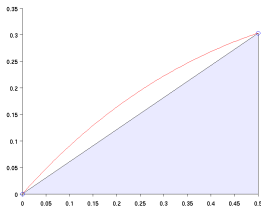
- ▶ But atoms and densities are much harder to obtain.

Numerical integration

- ▶ Equidistant atoms: **Newton-Cotes formulas**
 - Trapezoidal rule
 - Simpson rule
 - Romberg method (extrapolation of error)
- ▶ Optimal atoms: **Gaussian quadrature rules**

Basic trapezoidal rule and its error

Integral $\int_a^{a+h} f(x) dx$ so that f approximates by a linear function and calculate the area under the linear function or trapezoid.



$$p(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a)$$

Valid

$$\begin{aligned} \int_a^b f(x) dx &\approx \int_a^b p(x) dx = f(a)(b - a) + \frac{f(b) - f(a)}{b - a} \frac{(b - a)^2}{2} \\ &= \frac{(b - a)}{2} (f(a) + f(b)). \end{aligned}$$

Let

$$f[x, y] := \frac{f(y) - f(x)}{y - x},$$
$$f[x, y, z] := \frac{f[y, z] - f[x, y]}{z - x} = \frac{\frac{f(z) - f(y)}{y - z} - \frac{f(y) - f(x)}{y - x}}{z - x}.$$

The error here is:

$$\begin{aligned} E &= \int_a^b (f(x) - p(x)) dx = \int_a^b f[a, b, x](x - b)(x - a) dx \\ &= f[a, b, \xi] \cdot \int_a^{a+h} (x - b)(x - a) dx \\ &= \frac{f''(\eta)}{2} \left(-\frac{1}{6}(b - a)^3 \right) \\ &= \boxed{-\frac{(b - a)^3 f''(\eta)}{12}}, \end{aligned}$$

$\xi, \eta \in [a, b]$, the third equality follows by the mean value theorem and the fourth by a result on $f[a, b, \xi]$ -estimate.

Composite trapezoidal rule

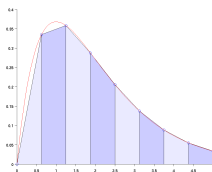
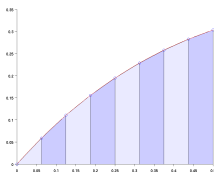
Unless the interval is very short, the obvious naive linear transformation usually does not give a good approximation of the integral.

If we divide the interval $[a, b]$ by the equidistant points $x_0, x_1, \dots, x_n$, i.e.,

$$h := h_i = x_{i+1} - x_i$$

is a constant and applying the basic trapezoidal rule on each interval, we get:

$$\int_a^b f(x) dx \approx \frac{h}{2} \sum_{i=0}^{n-1} f(x_i) + f(x_{i+1})$$
$$= \boxed{\frac{h}{2} (f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n))}$$



The error E_i on the interval $[x_i, x_{i+1}]$ is equal to

$$E_i = -\frac{h^3 \cdot f''(\eta_i)}{12} \quad \text{for some } \eta_i \in [x_i, x_{i+1}].$$

So the total error is

$$\begin{aligned} E &= \sum_{i=0}^{n-1} E_i = \sum_{i=0}^{n-1} -\frac{h^3 \cdot f''(\eta_i)}{12} = -n \cdot \frac{h^3 \cdot f''(\eta)}{12} \\ &= \boxed{-\frac{(b-a)h^2 \cdot f''(\eta)}{12}}, \end{aligned}$$

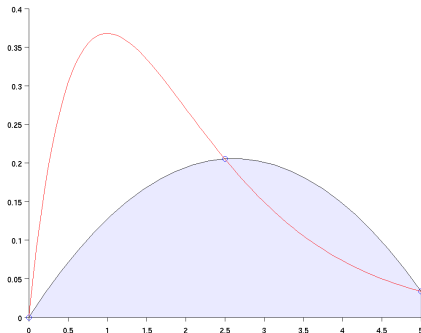
where $\eta \in [a, b]$ and we used the mean theorem in the third equality.

Algorithm: [click](#)

Basic Simpson's rule

Let p_2 be a polynomial of degree 2 with which we interpolate the points

$$(a, f(a)), \quad \left(\frac{a+b}{2}, f\left(\frac{a+b}{2}\right)\right), \quad (b, f(b)) :$$



$$p_2(x) = C_0 + C_1 \cdot (x - a) + C_2 \cdot (x - a) \left(x - \frac{a+b}{2} \right).$$

Multidimensional integration

We will separate the cases:

1. We are interested in

$$\int_{\Omega} f(x, y) \, dx \, dy,$$

where

$$\Omega = [a, b] \times [c, d] \subseteq \mathbb{R}^2.$$

2. We are interested in

$$\int_{\Omega} f(\underline{x}) \, d\mu,$$

where

$$\Omega \subseteq \mathbb{R}^d$$

is an arbitrary region in $\mathbb{R}^d$ and μ is a measure on Ω .

2-dimensional numerical integration

In the first case, we can use **two composite rules** for each variable separately.
Let the two sequences

$$a = x_0 < x_1 < x_2 < \dots < x_n = b$$

and

$$c = y_0 < y_1 < y_2 < \dots < y_n = d$$

divide the intervals $[a, b]$ and $[c, d]$ into n equal parts and $h = \frac{b-a}{n}$, $k = \frac{d-c}{n}$.
If we use the compound trapezoidal rules we get:

$$\begin{aligned} \int_{\Omega} f(x, y) \, dx \, dy &= \int_c^d \int_a^b f(x, y) \, dx \, dy = \int_c^d \left(\sum_{i=0}^{n-1} \frac{h}{2} (f(x_i, y) + f(x_{i+1}, y)) \right) dy \\ &= \frac{hk}{4} \sum_{j=0}^{n-1} \sum_{i=0}^{n-1} (f(x_i, y_j) + f(x_i, y_{j+1}) + f(x_{i+1}, y_j) + f(x_{i+1}, y_{j+1})) \end{aligned}$$

More compactly, we can say this by putting **multiply the same coefficients** in **table of function values**

$$\begin{pmatrix} f(x_n, y_0) & f(x_n, y_1) & f(x_n, y_2) & \cdots & f(x_n, y_{n-1}) & f(x_n, y_n) \\ f(x_{n-1}, y_0) & f(x_{n-1}, y_1) & f(x_{n-1}, y_2) & \cdots & f(x_{n-1}, y_{n-1}) & f(x_{n-1}, y_n) \\ \vdots & \cdots & \cdots & \cdots & \vdots & \\ f(x_1, y_0) & f(x_1, y_1) & f(x_1, y_2) & \cdots & f(x_1, y_{n-1}) & f(x_1, y_n) \\ f(x_0, y_0) & f(x_0, y_1) & f(x_0, y_2) & \cdots & f(x_0, y_{n-1}) & f(x_0, y_n) \end{pmatrix},$$

with **table of coefficients**

$$\frac{hk}{4} \cdot \begin{pmatrix} 1 & 2 & 2 & \cdots & 2 & 1 \\ 2 & 4 & 4 & 4 & 2 & 4 \\ \vdots & & & & & \vdots \\ 2 & 4 & 4 & \cdots & 4 & 2 \\ 1 & 2 & 2 & \cdots & 2 & 1 \end{pmatrix} = \frac{hk}{4} \begin{pmatrix} 1 \\ 2 \\ \vdots \\ 2 \\ 1 \end{pmatrix} (1 \quad 2 \quad \cdots \quad \cdots \quad 2 \quad 1)$$

and **sum all the values in the resulting array**.

d -dimensional Monte Carlo integration

- ▶ In the second case, we use **Monte Carlo methods**, which are based on the fact that

$$\int_{\Omega} f(x_1, \dots, x_d) d\mu = E_{\mu}(f(X_1, \dots, X_d)),$$

where $X = (X_1, \dots, X_d) : \Omega \rightarrow \Omega$ is the random vector, and E_{μ} is the expected value of the random variable $f(X_1, \dots, X_d)$.

- ▶ So basic idea is to **sample randomly** over Ω , and then **calculate the average of the function values**.
- ▶ For a sufficiently large number of random points, the average will be a good estimate of the value of the integral.
- ▶ The question of this part of the course will be **how to sample w.r.t. the given μ** .

The strong law of large numbers

Let $X_1, X_2, X_3, \dots$ be an infinite sequence of independent, identically distributed copies of a random variable X and

$$\bar{X}_n = \frac{1}{n}(X_1 + \dots + X_n)$$

be the empirical sample averages.

Theorem (Strong law of large numbers 1)

If $E[|X|] < \infty$, then

$\bar{X}_n$ converges almost surely to $E[X]$.

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ a measurable function and

$$\bar{f}(X)_n = \frac{1}{n}(f(X_1) + \dots + f(X_n)).$$

Theorem (Strong law of large numbers 2)

If $E[|f(X)|] < \infty$, then

$\bar{f}(X)_n$ converges almost surely to $E[f(X)]$.

The central limit theorem

Let $X_1, X_2, X_3, \dots$ be an infinite sequence of independent, identically distributed rv-s with finite mean μ and variance σ^2 . Let

$$Z_n = \frac{(\sum_{i=1}^n X_i) - n\mu}{\sqrt{n}\sigma}.$$

Theorem (Central limit theorem)

Z_n converges in distribution to $Z \sim N(0, 1)$.

It holds that

$$P(-2 < Z < 2) \approx 95.6\%.$$

Hence, for n large enough

$$P(-2 < Z_n < 2) \approx 95.6\%$$

or equivalently

$$P\left(-\frac{2\sigma}{\sqrt{n}} < \bar{X}_n - \mu < \frac{2\sigma}{\sqrt{n}}\right) \approx 95.6\%.$$

Corollary

Estimating $E[X]$ (resp. $E[f(X)]$) with $\bar{X}_n$ (resp. $E[\bar{f}_n]$) has error $O(n^{-1/2})$.

Monte Carlo integration

$$\int_{\Omega} f(\underline{x}) \underbrace{p(\underline{x})}_{\substack{\text{the density of} \\ \text{the distribution}}} d\underline{x} \approx \frac{1}{n} (f(x_1) + \dots + f(x_n)),$$

where n is large enough and $x_1, \dots, x_n$ are sampled independently w.r.t. the distribution $p(\underline{x})$.

Question: How to generate random sample w.r.t. $p(\underline{x})$?
The main focus of this course.

- ▶ Inverse method sampling.
Requires knowing inverse of the CDF F .
- ▶ Importance sampling.
Applicable when a 'similar' distribution, easier to sample from, is known.
- ▶ Rejection sampling.
Markov chains, Metropolis-Hastings algorithm.

Inverse method sampling

- ▶ **Linear congruential generators (LCG)** generate (pseudo)random numbers by

$$X_{n+1} = (aX_n + b) \mod m$$

for a, b, m large integers, e.g.,

$$a = 214013, b = 2531011, m = 2^{31}.$$

Some other triples (a, b, m) in use

https://en.wikipedia.org/wiki/Linear_congruential_generator.

- ▶ **Procedure.**

Input: $F : \mathbb{R} \rightarrow [0, 1]$... the CDF of a RV X .

Output: Independent sample $x_1, x_2, \dots$ from F .

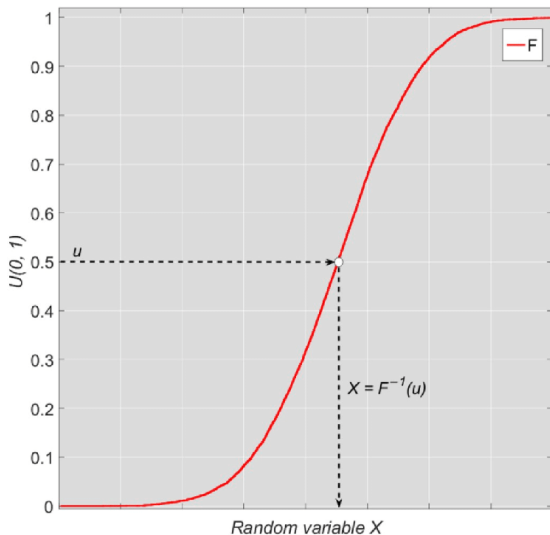
Step 1. Compute $F^{-1} : [0, 1] \rightarrow \mathbb{R}$.

Step 2. Sample uniformly $u_1, u_2, \dots$ on $[0, 1]$ using LCG.

Step 3. Compute $x_i = F^{-1}(u_i)$.

Inverse method sampling

Geometrically



Importance sampling

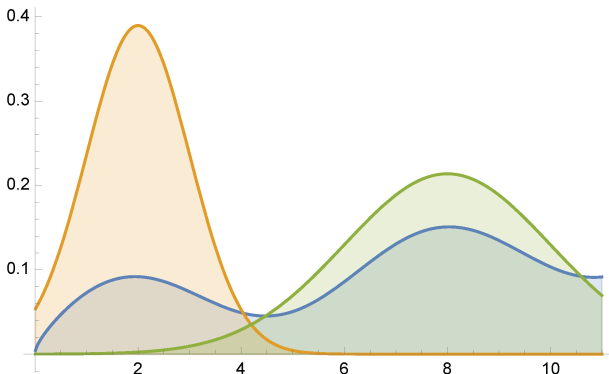
We have:

$$E_{p(x)dx}[f(X)] = \int f(x)p(x)dx = \int \frac{f(x)p(x)}{g(x)}g(x)dx = E_{g(x)dx}\left[\frac{f(x)p(x)}{g(x)}\right]$$

So instead of sampling from $p(x)$ one can sample from $g(x)$ and compute

$$\frac{1}{n} \sum_{i=1}^n \frac{f(x_i)p(x_i)}{g(x_i)}.$$

Problem: $\text{Var}\left[\frac{f(X)p(X)}{g(X)}\right]$ could be much bigger than $\text{Var}[f(X)]$.



Rejection sampling

Question: How to generate a sample from a uniform distribution on a disk?

First guess (which is wrong - why?) might be:

- ▶ Use polar coordinates

$$x = r \cos \varphi, \quad y = r \sin \varphi, \quad r \in [0, 1], \quad \varphi \in [0, 2\pi].$$

- ▶ Generate two samples:

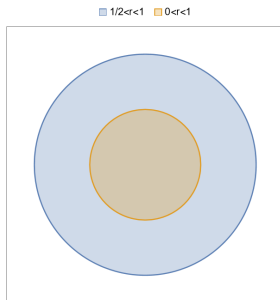
$$r_1, \dots, r_n \text{ from } U(0, 1) \quad \text{and} \quad \varphi_1, \dots, \varphi_n \text{ from } U(0, 2\pi).$$

- ▶ Compute

$$(x_i, y_i) = (r_i \cos \varphi_i, r_i \sin \varphi_i), \quad i = 1, \dots, n.$$

Rejection sampling

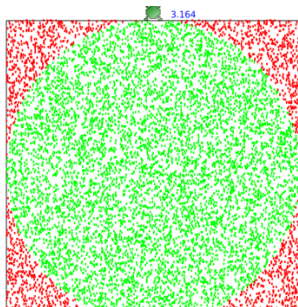
Question: what was wrong with the first guess above?



- If this was proper sampling, then the regions $(r, \varphi) \in [0, 1/2] \times [0, 2\pi)$ and $(r, \varphi) \in [1/2, 1] \times [0, 2\pi)$ should have the same area.

Rejection sampling

Question: What is the proper way of sampling uniformly from a disk B^2 ?



- Generate two samples:

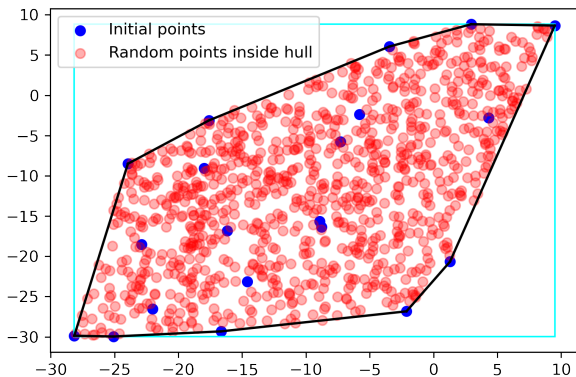
$x_1, \dots, x_n$ from $U(0, 1)$ and $y_1, \dots, y_n$ from $U(0, 1)$.

- If (x_i, y_i) is in the disk, **accept** it as a sample point. Otherwise **reject**.
- Except generating a sample from $U(B^2)$, you can also estimate **the area** of B^2 :

$$\text{Area}(B^2) = \frac{\#(\text{accepted points})}{\#(\text{all points})}.$$

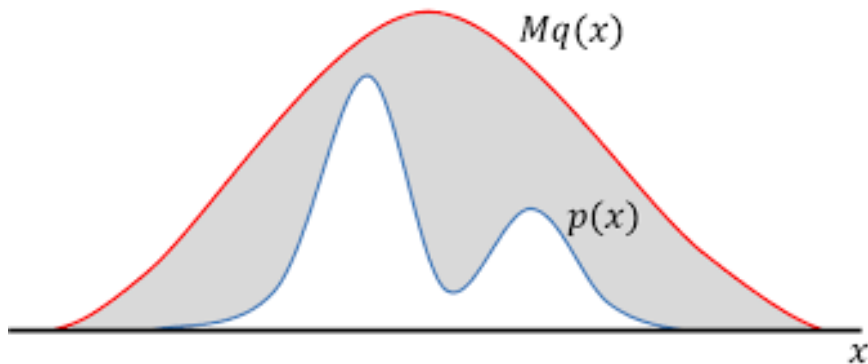
Rejection sampling

The same method can be used to sample uniformly from any other bounded region.



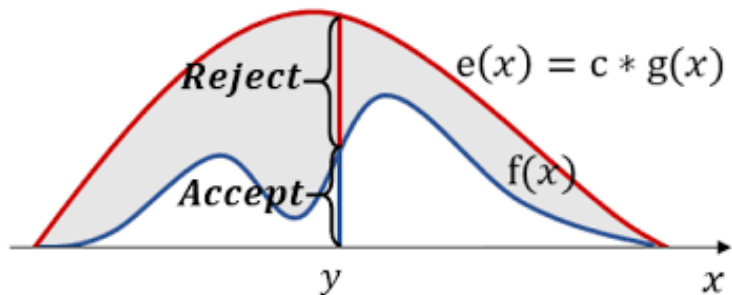
Rejection sampling

It can also be used to sample from a given distribution $p(x)$.



Rejection sampling

Sampling from $p(x)$



Procedure:

- ▶ **Step 1:** Choose a distribution $g(x)$ you know how to sample from.
- ▶ **Step 2:** Find $c > 0$ such that $cg(x) \geq p(x)$ for every x in the domain.
- ▶ **Step 3:** First sample x_i from $g(x)$ and then u_i from $U(0, cg(x_i))$.
 - ▶ Accept x_i if $u_i \leq p(x_i)$.
 - ▶ Otherwise reject x_i .

Rejection sampling

Sampling from $p(x)$

Remarks:

- ▶ Step 2 might be nontrivial.
- ▶ Instead of $p(x)$ one could also work with $dp(x)$ for some constant d . The final sample will still follow $p(x)$.

Markov chains

- ▶ Basic idea: Given a distribution $p(x)$ on a given state space S , generate a sequence $x_1, x_2, \dots$, such that for large n it holds that

$$P(x_n \in A) = \int_A p(x) dx \quad \text{where } A \subseteq S.$$

- ▶ **Restriction:** For the sake of simplicity (and to get intuition) we assume the state space S is **finite**.
- ▶ In the finite case we come into the domain of **linear algebra**; namely, **eigenpairs of entrywise nonnegative matrices** will play a crucial role.
- ▶ The results from the finite case **mostly generalize** to the **countably infinite** S under additional assumptions.
- ▶ Further on, some **further restrictions** are needed to generalize to **uncountably infinite** S .

Perron-Frobenius theorem

Important result from linear algebra

$\mathbb{R}^{n \times n}$... $n \times n$ matrices,

$\mathbb{R}_{\geq 0}^{n \times 1}, \mathbb{R}_{\geq 0}^{n \times n}$... $n \times 1$ vectors with nonnegative entries,

$\mathbb{R}_{> 0}^{n \times 1}, \mathbb{R}_{> 0}^{n \times n}$... $n \times 1$ vectors, $n \times n$ matrices with strictly positive entries

$A \in \mathbb{R}^{n \times n}$... a matrix

Spectral radius $\rho(A)$ is defined by

$$\rho(A) = \max \{ |\lambda| : \lambda \text{ is an eigenvalue of } A \}.$$

Theorem

Assume one of the following holds:

Perron, 1907: $A \in \mathbb{R}_{> 0}^{n \times n}$.

Frobenius, 1912: $A \in \mathbb{R}_{\geq 0}^{n \times n}$ and $A^m \in \mathbb{R}_{> 0}^{n \times n}$ for some $m \in \mathbb{N}$.

Then:

1. $\rho(A)$ is an **eigenvalue** of A with an eigenvector $\mathbf{v} \in \mathbb{R}_{> 0}^n$.
2. $\rho(A)$ is **the only** eigenvalue on a circle $\{z \in \mathbb{C} : |z| = \rho(A)\}$.
3. $\rho(A)$ has geometric (dimension of the corresponding eigenspace) and algebraic (multiplicity as a zero of characteristic polynomial) **multiplicity**

Perron-Frobenius theorem

Idea of the proof of Perron's theorem (pdf in the classroom)

Observation 1: For $\theta, \psi \in \mathbb{R}^n$ such that $\theta_i \geq \psi_i$ for each i and there is i_0 such that $\theta_{i_0} > \psi_{i_0}$, there exists $\epsilon > 0$ such that $A\theta > (1 + \epsilon)A\psi$. (*trivial*)

Observation 2: $\rho(A) = \lim_{n \rightarrow \infty} (\|A^n\|)^{1/n}$ (*Gelfand's formula, nontrivial*)

Proof: <https://planetmath.org/prooffogelfandspectralradiustheorem>

Observation 3: Let ψ be the eigenvector of an eigenvalue λ such that $|\lambda| = \rho(A)$. Then the vector $|\psi|$ with the components being absolute values of the components of ψ is an eigenvector of for $\rho(A)$. (*elementary proof, but uses GF*)

Observation 4: The only eigenvalue on $\{z \in \mathbb{C} : |z| = \rho(A)\}$ is $\rho(A)$. (*trivial*)

Extension to Perron-Frobenius case: Apply Peron's theorem to A^m and observe that $\sqrt[m]{\rho(A^m)} = \rho(A)$ with the same eigenvector.

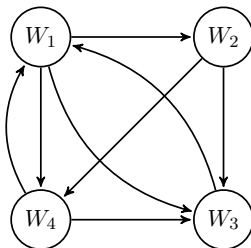
Directed adjacency graph $G(A)$

Let $A = [a_{ij}]_{i,j} \in \mathbb{R}^{n \times n}$ be a matrix.

A **directed adjacency graph** $G(A)$ has vertices $w_1, w_2, \dots, w_n$ and directed edges $w_i \rightarrow w_j$ for every i, j satisfying $a_{ij} \neq 0$.

Example

$$A = \begin{pmatrix} 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}$$



Condition $A^m \in \mathbb{R}_{>0}^{n \times n}$ in terms of graphs

Lemma

There exists a path of length m from vertex w_i to w_j if and only if $(A^m)_{ij} > 0$.

Idea of the proof. Induction on m .

Proposition

Let $A \in \mathbb{R}_{\geq 0}^{n \times n}$. Then there exists $m \in \mathbb{N}$ such that $A^m \in \mathbb{R}_{>0}^{n \times n}$ if and only if the following hold true:

- 1. Any vertex of $G(A)$ can be reached from any other vertex following some directed path.*
- 2. The greatest common divisor of the cycles in $G(A)$ is 1.*

Markov chains

$(S, \mathcal{S}, P)$... a measure space.

A **discrete stochastic process (DSP)** is a countable sequence $X(t_1), X(t_2), X(t_3), \dots$, where $t_1, t_2, t_3, \dots$ are different **times** and $X(t_i)$ are **random variables** in $(S, \mathcal{S})$. We will write $X_i := X(t_i)$.

We say that the DSP satisfies **the Markov property** if for every $i \in \mathbb{N}$ and every sequence $x_1, \dots, x_{i+1} \in S$ the following holds:

$$P(X_{i+1} = x_{i+1} | X_i = x_i, \dots, X_1 = x_1) = P(X_{i+1} = x_{i+1} | X_i = x_i).$$

In words: the conditional probability of going from state x_i to state x_{i+1} is **independent of the states $x_{i-1}, \dots, x_1$** .

Transition matrix and stationary distribution

We write $p_{ij} := P(X_{k+1} = x_j | X_k = x_i)$.

A **transition matrix** K of the DSP is a $\mathbb{N} \times \mathbb{N}$ matrix with the ij -th entry $K_{ij} = p_{ij}$.

Proposition

The sum of the entries of every row of K is equal to 1, i.e., $\sum_{j=1}^{\infty} K_{ij} = 1$ for every $i \in \mathbb{N}$.

We call a matrix with the property in Proposition, a **row stochastic matrix**.

Proposition

1. $K_{ij}^m = P(X_{k+m} = x_j | X_k = x_i)$.
2. Assuming a probability distribution $p \in \mathbb{R}^{|\mathcal{S}|}$, the probability distribution after one step of the DSP is pK .

We call a vector $\Pi = (p_1, \dots, p_{|\mathcal{S}|}) \in \mathbb{R}_{\geq 0}^{|\mathcal{S}|}$ a **stationary distribution** if

$$\Pi K = \Pi \quad \text{and} \quad \sum_{i=1}^{|\mathcal{S}|} p_i = 1$$

Stationary distribution and sampling

The idea of Markov chains for sampling from some distribution p : Construct a Markov chain with a stationary distribution p .

Question 1: Does a stationary distribution always exist?... Yes.

Question 2: Is a stationary distribution unique?... Not always.

Question 3: Does a DSP after a large number of steps converge to a/the stationary distribution?... Not always.

Stationary distribution

Existence

Lemma

Let $A \in \mathbb{R}_{\geq 0}^{n \times n}$ be a row stochastic matrix. Then $\rho(A) = 1$.

Proof.

- ▶ Let λ be an eigenvalue of A and $v \in \mathbb{R}^n$ a corresponding eigenvector. We have that:

$$\lambda v = Av = \left[\sum_{j=1}^n a_{ij} v_j \right]_{i=1}^n.$$

- ▶ Let $\|v\|_{\infty} := \max\{|v_1|, \dots, |v_n|\}$ be the supremum norm of a vector v . We have that

$$|[Av]_j| = \left| \sum_{j=1}^n a_{ij} v_j \right| \leq \sum_{j=1}^n a_{ij} |v_j| \leq \sum_{j=1}^n a_{ij} \|v\|_{\infty} = \|v\|_{\infty},$$

where in the first inequality a triangle inequality was used, in the second the definition of $|v_j| \leq \|v\|_{\infty}$ and in the third equality the fact that A is row stochastic.

- ▶ Take j_0 such that $|v_{j_0}| = \|v\|_{\infty}$. Then

$$|\lambda| \|v\|_{\infty} = |\lambda| |v_{j_0}| = |[Av]_{j_0}| \leq \|v\|_{\infty} \quad \Rightarrow \quad |\lambda| \leq 1.$$

- Since $A\vec{1} = \vec{1}$, where $\vec{1}$ is the vector of ones, 1 is an eigenvalue and $\rho(A) = 1$. ■

Corollary

Let $A \in \mathbb{R}_{\geq 0}^{n \times n}$ be a row stochastic matrix. Then $\rho(A^T) = 1$.

Proof. Use the fact that A and A^T have the same eigenvalues. This follows from the fact that

$$\det(A - \lambda I) = \det((A - \lambda I)^T) = \det(A^T - \lambda I). \quad \blacksquare$$

Stationary distribution

Uniqueness

Theorem

Let $A \in \mathbb{R}_{\geq 0}^{n \times n}$ be a row stochastic matrix and one of the following equivalent conditions hold:

1. $A^m \in \mathbb{R}_{> 0}^{n \times n}$ for some $m \in \mathbb{N}$.
2. The following holds:
 - 2.1 There is a path between each pair of vertices.
 - 2.2 The gcd of the lengths of all cycles is 1.

Then:

1. There is a unique vector $\Pi \in \mathbb{R}^{|S|}$ (up to multiplication with a constant) such that $\Pi A = \Pi$.
2. There is a unique vector $\Pi \in \mathbb{R}_{> 0}^{|S|}$ such that $\Pi A = \Pi$ and $\sum_i \Pi(i) = 1$.

Proof.

Use Perron-Frobenius theorem.



Irreducibility and aperiodicity

The previous theorem can be formulated in a more compact way once we introduce two properties of the Markov chain.

Let $\sim$ be the equivalence relation of the set S :

$w_i \sim w_j \iff$ There exist **directed paths in both directions**
between w_i, w_j in $G(A)$.

We call $\sim$ a **communication relation** and say that w_i and w_j **communicate**.

We call a Markov chain **irreducible**, if all pairs of states communicate.

We call an irreducible Markov chain **aperiodic**, if the gcd of the lengths of all cycles is 1.

Corollary

An irreducible, aperiodic Markov chain has a unique stationary distribution.

Ergodic theorem 1

Theorem

Let K be a transition matrix of an irreducible, aperiodic Markov chain with a finite state space $S = \{s_1, \dots, s_m\}$ and the unique stationary distribution Π . Let $f : S \rightarrow \mathbb{R}$ be a function. Then:

1. *Power iteration for Markov chains:*

$$\lim_{n \rightarrow \infty} K^n = \begin{pmatrix} \Pi \\ \vdots \\ \Pi \end{pmatrix}.$$

In particular, for any vector $v \in \mathbb{R}^m$ with $v_i \in [0, 1]$ and $\sum_i v_i = 1$ it holds true that

$$\lim_{n \rightarrow \infty} v^T K^n = \Pi.$$

2. *Strong law of large numbers for Markov chains:* Let $x_1, x_2, x_3, \dots$ be a sample generated by the Markov chain. Then:

$$\frac{1}{n} \sum_{i=1}^n f(x_i) \xrightarrow{\text{a.s.}} E_{\Pi}(f) = \int_S f(\underline{x}) \Pi(\underline{x}) d\underline{x} = \sum_{i=1}^m f(s_i) \Pi(s_i).$$

Remark. Note that the sample from 2. is not independent.

Central limit theorem for Markov chains

Theorem

Let K be a transition matrix of an irreducible, aperiodic Markov chain with a finite state space $S = \{s_1, \dots, s_m\}$ and the unique stationary distribution Π . Let $f : S \rightarrow \mathbb{R}$ be a function and $x_1, x_2, x_3, \dots$ be a sample generated by the Markov chain. Then:

$$\sqrt{n}(\bar{f}_n - E_{\Pi}[f(X)]) \underset{d}{\rightarrow} Y \sim N(0, \sigma^2),$$

where $\bar{f}_n = \frac{1}{n} \sum_{i=1}^n f(x_i)$ stands for the sample average and

$$\sigma^2 = \sum_{i,j=1}^n \text{Cov}(f(X_i), f(X_j)).$$

Convergence in practice: example 1

Let

$$K_1 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad K_2 = \begin{pmatrix} \frac{5}{100} & \frac{95}{100} \\ \frac{95}{100} & \frac{5}{100} \end{pmatrix}, \quad K_3 = \begin{pmatrix} \frac{95}{100} & \frac{5}{100} \\ \frac{5}{100} & \frac{95}{100} \end{pmatrix}$$

be 3 different transition matrices.

Since K_1, K_2, K_3 are row and column stochastic matrices (sums of rows and columns are all equal to 1), the unique stationary distribution for all three transition matrices is $\Pi = (1/2 \quad 1/2)$.

Question: Starting from a probability distribution vector $v = (v_1 \quad v_2)$, which K_i will converge to Π the quickest?

Studying the proof of the power iteration for Markov chains above we see, that the rate of convergence depends on the absolute value c of absolutely the second largest eigenvalues of the transition matrix, i.e., the smaller c implies the quickest convergence. **Answer:** For K_1 the second eigenvalue is

0, for K_2 , it is -0.9 and for K_3 it is 0.9 . K_1 converges already after one step, while K_2 and K_3 have the same rate of convergence.

Convergence in practice: example 2

Let K_1, K_2, K_3 be as above and

$$f : \{w_1, w_2\} \rightarrow \{5, -1\}$$

a function, defined by

$$f(w_1) = 5 \quad \text{and} \quad f(w_2) = -1.$$

We are interested in

$$\int_{\{w_1, w_2\}} f d\Pi = E_\Pi[f(X)] = \lim_n \frac{1}{n} (f(x_1) + \dots + f(x_n)), \quad (3)$$

where $x_1, x_2, \dots, x_n, \dots$ is a Markov chain generated by K_i . Clearly,

$$E_\Pi[f(X)] = \frac{1}{2}(5 - 1) = 2.$$

Question: Starting from a probability distribution vector $v = (v_1 \quad v_2)$, for which K_i the right hand side of (3) will converge to $E_\Pi[f(X)]$ the quickest?

Observing the variance in the CLT above we see, that the rate of convergence depends on $\sum_{i,j=1}^n \text{Cov}(f(X_i), f(X_j))$.

Convergence in practice: example 2

Observe that $\text{Cov}(f(X_i), f(X_j))$:

- ▶ For K_1 : If $i \neq j$, it is 0, since each next step is independent from the previous.
- ▶ For K_2 : If i and j are at odd distance, it is negative, since there is a huge tendency to change the state.
- ▶ For K_3 : For every i, j , it is positive, since there is a huge tendency to remain in the same state.

Answer to the question: K_2 has the quickest rate of convergence, since σ^2 is the smallest due to negative lag- k autocovariances for odd k . K_3 converges the slowest.

Algorithm: [click](#)

Monte Carlo Markov chains

Basic idea

Input:

Ω ... a subset of $\subseteq \mathbb{R}^d$,
 p ... a density of some distribution on Ω ,
 $f : \Omega \rightarrow \mathbb{R}$... a function.

Output: Approximation of $E_p(f) = \int_{\Omega} f p(\underline{x}) d\underline{x}$ using the CLT for Markov chains based on the sample generated.

Procedure:

- Step 1:** Construct a Markov chain (MCh) with a stationary distribution p .
Do it so that it reaches p as quick as possible.
- Step 2:** Generate a long sequence $x_1, x_2, \dots, x_n$ from this MCh.
Ignore a certain number of starting elements of the chain.
- Step 3:** Compute $E_p \approx \frac{1}{n} \sum_{i=1}^n f(x_i)$.

A detailed balance condition

Clearly, exact MCh cannot be generated. Iteratively improve the guess at each step.

We say Π satisfies a **detailed balance (DB)** condition if for every $x, y \in S$ it holds true that

$$\Pi(y)K(y, x) = \Pi(x)K(x, y).$$

Proposition

A Π which satisfies a DB condition is stationary.

Proof.

$$\begin{aligned}\Pi K &= \left(\sum_x \Pi(x) K(x, y) \right)_y = \left(\sum_x \Pi(y) K(y, x) \right)_y \\ &= \left(\Pi(y) \sum_x K(y, x) \right)_y = \left(\Pi(y) \right)_y = \Pi.\end{aligned}$$

In the computation above the sum $\sum_x$ goes over all states x , while $(\cdot)_y$ denotes the entry of the vector corresponding to state y . The first equality is definition of vector-to-matrix multiplication, then in the second we used DB condition, while in the fourth we used that K is row stochastic. $\square$

Construction of the Markov chain with a desired stationary distribution Π

1. Choose some transition matrix K (called **proposal matrix**) defining a Markov chain.
2. Do each step according to K but accept/reject the move from a **current state** x to a **new state** y according to the ratio

$$\alpha(x, y) := \frac{\Pi(y)K(y, x)}{\Pi(x)K(x, y)}.$$

The **bigger** this **ratio**, the **bigger** the **tendency to move**.

Indeed, to achieve the detailed balanced condition we want that $\alpha(x, y)$ is as close to 1 as possible. Actually, we are following the MCh with a transition matrix

$$\left(\tilde{K}_{ij}\right)_{ij} := \left(\alpha_{ij}K_{ij}\right)_{ij}.$$

Metropolis-Hastings algorithm

Input:

- p ... a density of some distribution on $\Omega \subseteq \mathbb{R}^d$,
- $K(x, y)$... a proposal distribution for MCh,
- x_0 ... an initial state,
- N ... the number of steps.

Output: A sample of N points (not necessarily different) from p .

Procedure:

For $i = 1, \dots$ repeat the following steps:

1. Sample x^* from the distribution $K(x_i, y)$.
2. Compute $\alpha(x_{i-1}, x^*) = \min\left(\frac{p(x^*)K(x^*, x_{i-1})}{p(x_{i-1})K(x_{i-1}, x^*)}, 1\right)$.
3. Sample u from $U(0, 1)$.
4. If $u \leq \alpha(x_{i-1}, x^*)$, $x_i = x^*$. (accept)
5. Else $u > \alpha(x_{i-1}, x^*)$, $x_i = x_{i-1}$. (reject)

Remark: If $K(x, y) = K(y, x)$ for every x, y , then $\alpha(x_{i-1}, x^*) = \frac{p(x^*)}{p(x_{i-1})}$ in step 2 above and we get a **Metropolis algorithm**.

Metropolis-Hastings algorithm: examples

Algorithm: [click](#)

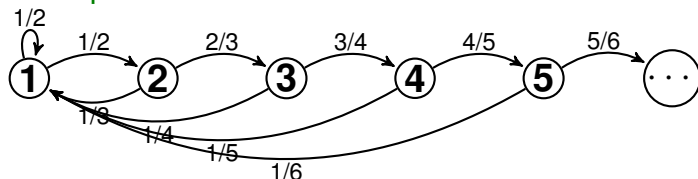
Generalization to countable Markov chains

Let S be a countable state space and $X_0, X_1, X_2, \dots$ a MCh. For $x \in S$ let

$$T_x = \min\{n \geq 1 : X_n = x | X_0 = x\}.$$

A random variable T_x is called a **first return time**.

Example



Proposition

$P(T_1 < \infty) = 1$ and $E[T_1] = \infty$.

Types of states

We separate the following types of states:

1. **Transient:** If $P(T_x < \infty) < 1$.
2. **Recurrent:** If $P(T_x < \infty) = 1$.
 - 2.1 **Positive recurrent:** $E[T_x] < \infty$.
 - 2.2 **Null recurrent:** $E[T_x] = \infty$.

Theorem

1. All states of an irreducible MCh are of the **same type**.
2. A positive recurrent MCh has a **unique stationary distribution**:

$$\pi(x) = \frac{1}{E[T_x]} \quad \forall y \in S.$$

3. Every finite irreducible MCh is **positive recurrent**.